Controlling $2+2\to4$ Circuit Formation in a Fully Trainable Two-Block ReLU Network

Summary. We derive a zero-temperature stationary phase diagram for a two-block residual ReLU network with independent weight decays. The calculation has two nested levels. At fixed representation strengths (s, q), we analytically eliminate the parent readouts, target-producing weights, and $d4$ readout. We then minimize the resulting scalar free energy over (s, q) separately for a direct and a compositional circuit family, and select the family with lower minimum. The second block uses a compositional $2+2 \to 4$ gate when (i) its self-consistent representation satisfies $Q_2/Q_1^{(1)} > 1/\sqrt{8}$, (ii) the compositional free energy is below the direct one, and (iii) decay does not suppress the target. Layerwise decay can control the compositional share, but it creates a tradeoff: increasing first-block decay suppresses the unwanted first-block $d4$ shortcut and also suppresses the wanted $d2$ parents. A compositional phase therefore exists only inside a finite decay window.

1. Model and assumptions

For target parity $\chi_a(x) \in \{-1, +1\}$, let $f_a(x) = w_a^\top h_2(x)$ and define its target-mode amplitude $\widehat{A}_a = \mathbb{E}_x[f_a(x)\chi_a(x)]$. Under the pure aligned-mode ansatz $f_a = \widehat{A}_a\chi_a$, the sample loss reduces as $\mathbb{E}_x[(f_a - \chi_a)^2] = (1 - \widehat{A}_a)^2$, since $\chi_a^2 = 1$. The network and reduced weighted objective are therefore

$$\begin{aligned} &h_1 = h_0 + \sigma(V_1 h_0), \qquad h_2 = h_1 + \sigma(V_2 h_1), \\ &f_a = w_a^\top h_2, \\ &\mathcal{J} = \frac{\gamma_2}{2} \sum_{a\in\{12,34\}} (1 - \widehat{A}_a)^2 + \frac{\gamma_4}{2}(1 - \widehat{A}_4)^2 + \frac{1}{2}\big(\lambda_E\|E\|^2 + \lambda_1\|V_1\|^2 + \lambda_2\|V_2\|^2 \\ &\qquad\qquad + \lambda_p\|W_{d2}\|^2 + \lambda_4\|w_4\|^2\big)\,. \end{aligned} \tag{1}$$

The derivation assumes: (A1) the full four-bit cube and pure ReLU; (A2) four equal-norm, orthogonal singleton Walsh vectors of norm $s = Q_1^{(0)}$; (A3) two equal-norm, orthogonal parent vectors q_{12}, q_{34} of norm $q = Q_2$; (A4) balanced rank-one gates, sufficient width, and additive parameter norms for parent and target sectors; (A5) block-1 and block-2 target vectors align with their common readout; (A6) no pre-existing frozen $d4$ feature; and (A7) the global zero-temperature minimum is reached. Thus this is an equilibrium mode theory, not a claim about finite-time SGD or dynamical lock-in.

The symbol V_1 in Eq. (1) is the *entire* first-block weight matrix. Under the additive-sector assumption A4, we decompose it conceptually as

$$V_1 = V_{1,\text{parent}} + V_{1,d4} + V_{1,\text{other}}, \qquad \|V_1\|_F^2 = \|V_{1,\text{parent}}\|_F^2 + \|V_{1,d4}\|_F^2 + \|V_{1,\text{other}}\|_F^2. \tag{2}$$

$V_{1,\text{parent}}$ is not a separate network parameter. It denotes the minimum-norm component of V_1 that implements the two supervised $1+1 \to 2$ parent gates. The component $V_{1,d4}$ implements the direct block-1 target route D_1 introduced below. Their costs can be minimized separately only because A4 assumes orthogonal sectors and additive squared norms.

The singleton byproduct of an ideal parent gate is not a fitted parameter:

$$\sigma[a(x_1 + x_2)] = \frac{a}{2}(1 + x_1 + x_2 + x_1x_2). \tag{3}$$

It has the same singleton and pair coefficient. If this generated singleton vector is orthogonal to the residual singleton vector, then

$$Q_1^{(1)} = \sqrt{s^2 + q^2}. \tag{4}$$

For overlap ρ, replace $s^2 + q^2$ by $s^2 + q^2 + 2\rho sq$. All gate-selection statements below remain exact when expressed using the measured $Q_1^{(1)}$.

2. Derivation roadmap: what is minimized, and in what order

The calculation is an algebraic elimination of weights from one simultaneous objective, not a claim that training executes the following steps in time. For each candidate target family $r \in \{H_D, H_C\}$, the minimization is organized as follows.

1. Choose nonnegative order parameters (s, q), where s fixes the four singleton-feature norms and q fixes the two $d2$ parent-feature norms.

2. Holding (s, q) fixed, minimize the $d2$ losses and the penalties on E, $V_{1,\text{parent}}$, and W_{d2}. This gives the parent-sector free energy $\Phi(s, q)$.

3. Still holding (s, q) fixed, choose either the direct hybrid $H_D = D_1 + D_2$ or the compositional hybrid $H_C = D_1 + C$. Minimize its hidden weights and the shared $d4$ readout at fixed total target amplitude A, then minimize over $A \geq 0$. This gives the target-sector free energy $G(\zeta_r)$.

4. Finally minimize $\Phi(s, q) + G(\zeta_r)$ over (s, q). The two families perform this outer minimization separately, so their optimal representations need not be the same. The stationary phase is the family with lower final minimum.

Schematically, the nested calculation is

$$F_r = \min_{s,q\geq 0} \left[\underbrace{\min_{B,\, E, V_{1,\text{parent}}, W_{d2}} \mathcal{J}_{d2}}_{\Phi(s,q)} + \underbrace{\min_{A\geq 0,\, \theta_r, w_4} \mathcal{J}_{d4}}_{G(\zeta_r)} \right], \tag{5}$$

where B is the common fitted amplitude of the two parent targets and θ_r denotes the target-producing hidden weights in family r.

3. Eliminating weights: the two stationary free energies

Equation (1) specifies the penalties but does not by itself imply the norms below; they are the minimum-norm realizations under A1–A4. Write e_i for the four orthogonal singleton vectors. Then $\|E\|_F^2 = \sum_i \|e_i\|^2 = 4s^2$. For one balanced $1+1 \to 2$ gate of hidden norm u, the preactivation amplitude per parent is $us/\sqrt{2}$; Eq. (3) therefore gives $q = us/(2\sqrt{2})$, or $u^2 = 8q^2/s^2$. The two disjoint parent gates add in squared norm. Finally, the minimum-norm readout satisfying $w_{ij}^\top q_{ij} = B$ is $w_{ij} = Bq_{ij}/q^2$, with $\|w_{ij}\|^2 = B^2/q^2$. Thus the two readout rows also add, giving

$$\|E\|^2 = 4s^2, \qquad \|V_{1,\text{parent}}\|^2 = \frac{16q^2}{s^2}, \qquad \|W_{d2}\|^2 = \frac{2B^2}{q^2}. \tag{6}$$

At fixed (s, q), the parent-sector objective is therefore

$$\Phi(s, q) \equiv \min_{B\geq 0} \left\{ \gamma_2(1 - B)^2 + 2\lambda_E s^2 + 8\lambda_1 \frac{q^2}{s^2} + \lambda_p \frac{B^2}{q^2} \right\}. \tag{7}$$

This definition includes both $d2$ losses and the regularization needed to create and read out the singleton and parent representations. It excludes every $d4$-specific weight and the $d4$ loss. Minimizing the displayed expression over the common parent amplitude gives

$$\begin{aligned} B^*(q) &= \frac{\gamma_2 q^2}{\gamma_2 q^2 + \lambda_p}, \\ \Phi(s, q) &= 2\lambda_E s^2 + 8\lambda_1 \frac{q^2}{s^2} + \frac{\gamma_2 \lambda_p}{\gamma_2 q^2 + \lambda_p}. \end{aligned} \tag{8}$$

There are three target gates: D_1 is direct in block 1, D_2 is direct in block 2, and C is compositional in block 2. Their efficiencies per hidden-weight norm are

$$\eta_{D_1} = \frac{s}{8}, \qquad \eta_{D_2} = \frac{\sqrt{s^2 + q^2}}{8}, \qquad \eta_C = \frac{q}{2\sqrt{2}}. \tag{9}$$

The unrestricted competitors are the aligned hybrids $H_D = D_1 + D_2$ and $H_C = D_1 + C$. It is useful to define the inverse hidden-cost factors

$$K_1 = \frac{\eta_{D_1}^2}{\lambda_1} = \frac{s^2}{64\lambda_1}, \quad K_{D_2} = \frac{\eta_{D_2}^2}{\lambda_2} = \frac{s^2 + q^2}{64\lambda_2}, \quad K_C = \frac{\eta_C^2}{\lambda_2} = \frac{q^2}{8\lambda_2}. \tag{10}$$

Larger K means that a route can produce more target amplitude per unit hidden-layer penalty.

Here is the explicit meaning of "balancing the hidden routes against the readout." For family r, let $\mathcal{I}_D = \{D_1, D_2\}$ or $\mathcal{I}_C = \{D_1, C\}$. Let $u_i \geq 0$ be the norm of the target-specific hidden weights assigned to route i, and let $t = \|w_4\|$ be the

norm of their shared, aligned readout. Route i creates hidden target amplitude $\eta_i u_i$, so the total output amplitude is $A = t\sum_{i\in\mathcal{I}_r}\eta_i u_i$. At fixed A, the target-sector regularization is therefore the constrained minimum

$$\mathcal{R}_r(A) = \min_{t,u_i\geq 0}\left\{\frac{1}{2}\sum_{i\in\mathcal{I}_r}\lambda_i u_i^2 + \frac{\lambda_4}{2}t^2 \;:\; A = t\sum_{i\in\mathcal{I}_r}\eta_i u_i\right\}, \tag{11}$$

where $\lambda_{D_1} = \lambda_1$ and $\lambda_{D_2} = \lambda_C = \lambda_2$. Define $K_r = \sum_{i\in\mathcal{I}_r}\eta_i^2/\lambda_i$. For fixed t, the least-cost allocation is

$$u_i = \frac{A}{tK_r}\frac{\eta_i}{\lambda_i}, \qquad \mathcal{R}_r(A;t) = \frac{A^2}{2t^2K_r} + \frac{\lambda_4 t^2}{2}. \tag{12}$$

The first term decreases when the readout is enlarged, while the second increases. Minimizing over t makes the two terms equal and gives

$$t^4 = \frac{A^2}{\lambda_4 K_r}, \qquad \mathcal{R}_r(A) = A\sqrt{\frac{\lambda_4}{K_r}} = \frac{A}{\zeta_r}, qquad\zeta_r^2 \equiv \frac{K_r}{\lambda_4}. \tag{13}$$

This is the "balance": at the optimum, the hidden-route penalty and readout penalty are equal. It also fixes how the output amplitude is divided among routes:

$$\frac{A_i}{A} = \frac{t\eta_i u_i}{A} = \frac{\eta_i^2/\lambda_i}{K_r} = \frac{K_i}{K_r}. \tag{14}$$

For the two candidate families, the effective inverse-cost efficiencies are consequently

$$\zeta_D^2 = \frac{K_1 + K_{D_2}}{\lambda_4} = \frac{1}{\lambda_4}\left[\frac{s^2}{64\lambda_1} + \frac{s^2+q^2}{64\lambda_2}\right], \qquad \zeta_C^2 = \frac{K_1 + K_C}{\lambda_4} = \frac{1}{\lambda_4}\left[\frac{s^2}{64\lambda_1} + \frac{q^2}{8\lambda_2}\right]. \tag{15}$$

After this hidden/readout minimization, only the scalar target-amplitude minimization remains:

$$G(\zeta_r) \equiv \min_{A\geq 0}\left\{\frac{\gamma_4}{2}(1-A)^2 + \frac{A}{\zeta_r}\right\}. \tag{16}$$

Solving it gives

$$\begin{aligned} A^*(\zeta) &= \left[1 - \frac{1}{\gamma_4\zeta}\right]_+, \\ G(\zeta) &= \begin{cases} \zeta^{-1} - (2\gamma_4\zeta^2)^{-1}, & \gamma_4\zeta > 1, \\ \gamma_4/2, & \gamma_4\zeta \leq 1. \end{cases} \end{aligned} \tag{17}$$

The closed phase theory is therefore

$$\boxed{F_D = \min_{s,q\geq 0}\{\Phi(s,q) + G(\zeta_D)\}, \qquad F_C = \min_{s,q\geq 0}\{\Phi(s,q) + G(\zeta_C)\}.} \tag{18}$$

Each phase chooses its own representation; the joint-training transition $F_C = F_D$ can therefore be first order even though the fixed-representation crossover is simple.

4. Phase diagram and control laws

At fixed (s,q) the common D_1 term cancels, as do λ_2 and λ_4:

$$\zeta_C > \zeta_D \quad\Longleftrightarrow\quad \frac{q}{\sqrt{s^2+q^2}} = \frac{Q_2}{Q_1^{(1)}} > \frac{1}{\sqrt{8}}. \tag{19}$$

Thus layerwise decay does not directly move the block-2 gate boundary. It moves the self-consistent (s,q) and controls how much output goes through block 1.

To define the compositional share, write the fitted target amplitude in the active H_C phase as $A = A_{D_1} + A_C$, where A_{D_1} is supplied by the direct block-1 route and A_C by the compositional block-2 route. We define

$$f_C \equiv \frac{A_C}{A_{D_1} + A_C}. \tag{20}$$

Thus f_C is a share of the signed fitted $d4$ amplitude under the aligned positive-amplitude ansatz; it is not a fraction of neurons or a probability. Applying Eq. (14) with $K_r = K_1 + K_C$ gives

$$\boxed{f_C = \frac{q^2/(8\lambda_2)}{s^2/(64\lambda_1) + q^2/(8\lambda_2)} = \frac{8\lambda_1 q^2}{\lambda_2 s^2 + 8\lambda_1 q^2}.} \tag{21}$$

The number $p \in (0,1)$ is now an externally chosen *purity target*: for example, $p = 1/2$ asks that composition provide a majority of the fitted $d4$ amplitude, whereas $p = 0.9$ asks for a 90% compositional circuit. It is not another stationarity condition. Imposing the design requirement $f_C \geq p$ and rearranging Eq. (21) gives

$$\begin{aligned} f_C \geq p \quad &\Longleftrightarrow \quad K_C \geq \frac{p}{1-p} K_1 \\ &\Longleftrightarrow \quad \frac{\lambda_1}{\lambda_2} \geq \frac{p}{1-p} \frac{s^2}{8q^2}. \end{aligned} \tag{22}$$

But Eq. (8) shows that increasing λ_1 simultaneously makes the parents more expensive. The desired circuit therefore requires a window

$$\boxed{\lambda_{1,\text{purity}} < \lambda_1 < \lambda_{1,\text{parent collapse}}, \quad F_C < F_D, \quad \gamma_4 \zeta_C > 1.} \tag{23}$$

Here $\lambda_{1,\text{purity}}$ is the fixed-(s,q) lower bound from Eq. (22), evaluated at the compositional stationary solution, while $\lambda_{1,\text{parent collapse}}$ denotes the point at which the outer minimization of Eq. (18) moves to $q = 0$, a parents-only solution, or a task-suppressed solution. The remaining two conditions say that the compositional family must beat the direct family globally and that its target amplitude $A^*(\zeta_C)$ must be nonzero.

These are distinct tests: Eq. (19) chooses the block-2 gate at fixed (s,q); $F_C < F_D$ chooses the global family after re-optimizing (s,q); $\gamma_4 \zeta_C > 1$ makes the $d4$ target active; and $f_C \geq p$ is the analyst's optional purity threshold.

How to tune the fully trainable model. (1) Decay w_4; without it, hidden/readout rescaling invalidates norm comparison. (2) Preserve the parents with strong $d2$ supervision and modest $\lambda_E, \lambda_1, \lambda_p$ until the *final* measured $Q_2/Q_1^{(1)}$ exceeds $1/\sqrt{8}$. (3) Lower λ_2 relative to λ_1 to move target amplitude from D_1 into block 2, then increase λ_1/λ_2 only while the parents remain supercritical. (4) Measure separately the signed $d4$ readout contributions of h_1 and $\sigma(V_2 h_1)$; block-2 neuron classification alone does not measure total circuit share. (5) If no interval satisfies Eq. (23), whole-layer decay is insufficient: separate parent-producing and target-specific directions (soft subspace pinning or a decayed $d4$ adapter). All weights may remain trainable; freezing is not theoretically required.

Control	Desired movement	Limiting failure
$\lambda_1\uparrow/\lambda_2$	$f_C \uparrow$; suppress D_1	parent collapse: $q \downarrow$
$\lambda_2\downarrow$	make block-2 gates cheaper	large block-2 norms
$\lambda_p\downarrow$ or $\gamma_2\uparrow$	preserve $d2$ parents	weak parent regularization
$\lambda_4 > 0$	fixes hidden/readout scale	target suppression if too large
$\lambda_E\downarrow$	preserve singleton sources	embedding norms grow
$\gamma_4\uparrow$	avoids $A^* = 0$	does not by itself select C

Scope. Nonzero singleton overlap ρ, unequal parent norms, imperfect target alignment, fixed block-1 $d4$ leakage, or multifunctional neurons enlarge the order-parameter set but do not change the measured local boundary in Eq. (19). Finite-temperature Langevin dynamics replaces $F_C = F_D$ by $F_C - F_D + \frac{T}{2}\log(\det' H_C/\det' H_D) - T\log(g_C/g_D) = 0$ and rounds the transition. Ordinary full-batch SGD is effectively zero-noise but need not reach the global saddle, so metastability and training history must be checked experimentally.

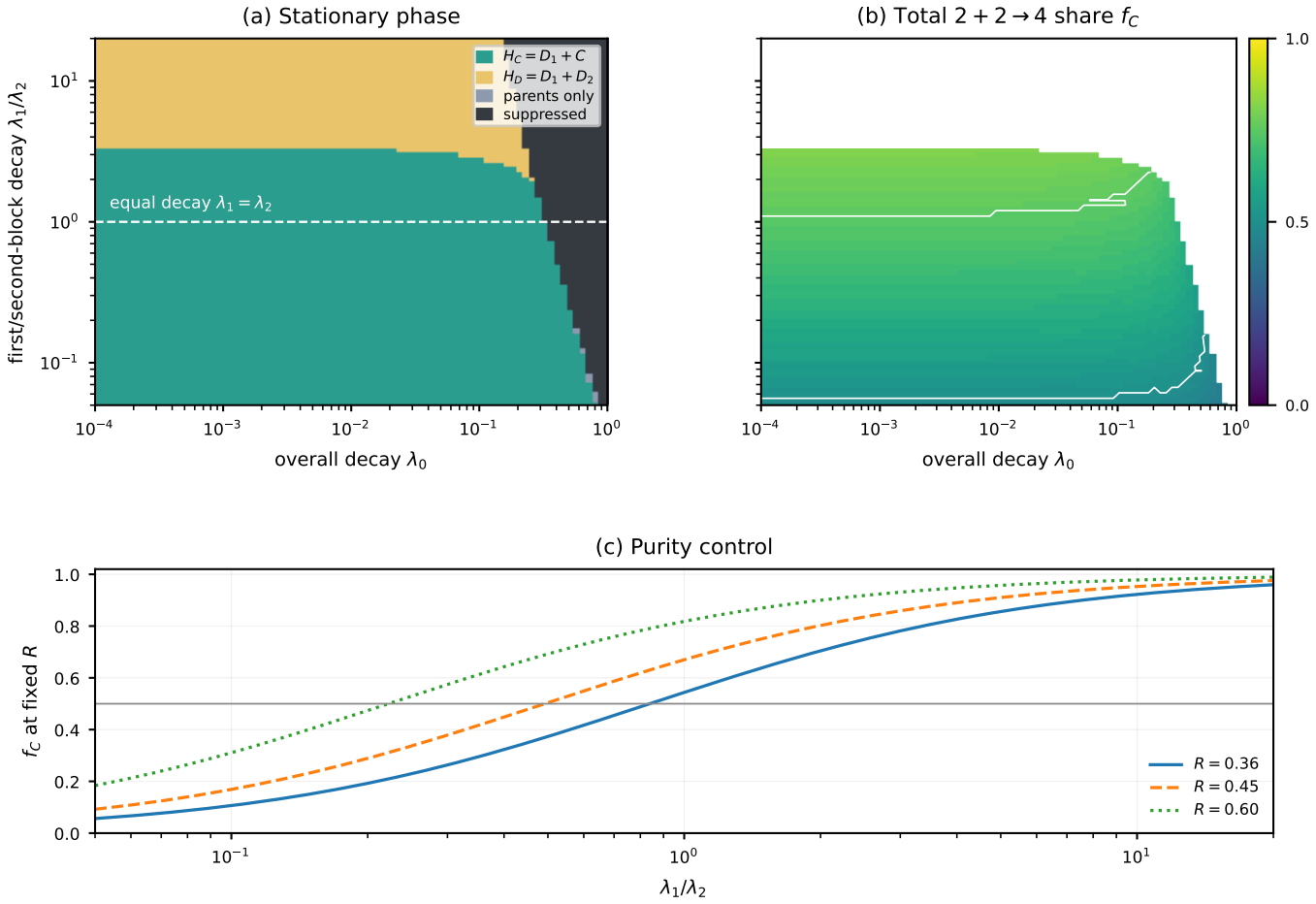


Figure 1: **Canonical stationary phase diagram.** We set $\gamma_2 = \gamma_4 = 2$, $\lambda_E = \lambda_p = \lambda_4 = \lambda_2 = \lambda_0$, λ_1/λ_2 as the vertical control, and use assumptions A1–A7. (a) Global minima of Eq. (18). Too much first-block decay destroys the parents and selects H_D; sufficiently large overall decay suppresses the tasks. (b) Composition can be selected yet remain impure when D_1 is cheap; white contours mark the target-amplitude shares $f_C = 0.5, 0.75, 0.9$. (c) Eq. (21) at fixed measured accessibility $R = Q_2/Q_1^{(1)}$: increasing λ_1/λ_2 suppresses the block-1 shortcut, provided the representation does not fall below Eq. (19).